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Antenna module

Abstract

An antenna module includes a first antenna, a second antenna, a first ground plane, and a second ground plane. The first antenna is used to excite at a first frequency band. The second antenna is disposed beside the first antenna and includes a first radiator with a feeding end, a second radiator connected to the first radiator, and a third radiator connected to the first radiator and including a ground end. The first radiator excites at a second frequency band, and a part of the first radiator and the second radiator excite at a third frequency band. The first ground plane is located below the first antenna and spaced apart from the first antenna. The second ground plane is located below the second antenna and spaced apart from the first ground plane. The ground end is connected to the second ground plane.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 111115609, filed on Apr. 25, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technology Field

(2) The disclosure relates to an antenna module, and in particular, to an antenna module having dual antennas.

Description of Related Art

(3) At present, the antenna module used in the bicycle tracker (tracker) includes a GPS antenna and an LTE antenna. The bicycle tracker may perform location positioning with GPS satellites in outer space through the GPS antenna, and then transmit the location positioning data to the mobile phone and the cloud server through the LTE antenna, which enables people to track the location of the bicycle immediately.

(4) However, in light of the bicycle tracker pursuing a light, thin, short, and small appearance, the GPS antenna is disposed very close to the LTE antenna, which leads to poor isolation between the

two antennas, thereby interfering with each other and resulting in poor antenna reception.

SUMMARY

(5) The disclosure provides an antenna module with favorable isolation between two antennas.

(6) An antenna module of the disclosure includes a first antenna, a second antenna, a first ground plane, and a second ground plane. The first antenna is for exciting at a first frequency band. The second antenna is disposed beside the first antenna, and includes a first radiator with a feeding end, a second radiator connected to the first radiator, and a third radiator connected to the first radiator and including a ground end. The first radiator excites at a second frequency band, and a part of the first radiator and the second radiator excite at a third frequency band. The first ground plane is located below the first antenna and spaced apart from the first antenna. The second ground plane is located below the second antenna and spaced apart from the first ground plane. The ground end is connected to the second ground plane.

(7) In an embodiment of the disclosure, the antenna module above further includes a conductor spacer connected to the second ground plane and extending from the second ground plane toward the second antenna. The conductor spacer is located between the first antenna and a part of the second antenna and separated from the first antenna and the second antenna.

(8) In an embodiment of the disclosure, the above-mentioned antenna module further includes a third ground plane located below the second ground plane and connected to the second ground plane through multiple via holes.

(9) In an embodiment of the disclosure, the above-mentioned first ground plane is insulated from the third ground plane.

(10) In an embodiment of the disclosure, the above-mentioned first ground plane and the second ground plane are coplanar, and an area of the second ground plane is greater than an area of the first ground plane.

(11) In an embodiment of the disclosure, a distance between the above-mentioned first ground plane and the second ground plane is between 0.5 mm and 1.5 mm.

(12) In an embodiment of the disclosure, a projection of the above-mentioned first antenna to the plane where the first ground plane is located lies within the range of the first ground plane, and a projection of the second antenna to the plane where the second ground plane is located lies within the range of the second ground plane.

(13) In an embodiment of the disclosure, the above-mentioned second antenna surrounds multiple sides of the first antenna, and the second ground plane surrounds multiple sides of the first ground plane.

(14) In an embodiment of the disclosure, a length of the above-mentioned first radiator is $\frac{1}{4}$ wavelength of the second frequency band, and lengths of a part of the first radiator and the second radiator are $\frac{1}{4}$ wavelength of the third frequency band.

(15) In an embodiment of the disclosure, a side length of the above-mentioned first antenna is $\frac{1}{2}$ wavelength of the first frequency band.

(16) Based on the above, the antenna module of the disclosure has the first antenna for exciting at the first frequency band and the second antenna for exciting at the second frequency band and the third frequency band so as to provide multiple applications. When the two antennas of the conventional structure are too close, the isolation is poor. In the antenna module of the disclosure, the first ground plane is located below the first antenna and spaced apart from the first antenna. The second ground plane is located below the second antenna and spaced apart from the first ground plane. The ground end of the second antenna is connected to the second ground plane. Through the design of the first antenna and the second antenna having the corresponding first ground plane and second ground plane separately, the antenna module of the disclosure may enable the first antenna and the second antenna to have more favorable isolation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of an antenna module according to an embodiment of the disclosure.
- (2) FIG. 2 is a schematic diagram of the antenna module shown in FIG. 1 with the insulating support, the first antenna, and the second antenna hidden.
- (3) FIG. 3 is a frequency-S11 relation diagram of the first antenna and the second antenna of the antenna module shown in FIG. 1.
- (4) FIG. 4 is a frequency-isolation relation diagram between the conventional antenna and the antenna module shown in FIG. 1.

DESCRIPTION OF THE EMBODIMENTS

- (5) FIG. 1 is a schematic diagram of an antenna module according to an embodiment of the disclosure. Referring to FIG. 1, an antenna module **100** of the embodiment is, for example, an antenna module applied to a bicycle tracker (tracker), but the application of the antenna module **100** is not limited thereto. The antenna module **100** of the embodiment includes a first antenna **110**, a second antenna **120**, a first ground plane **130**, and a second ground plane **132**. The first antenna **110** is, for example, a GPS ceramic antenna, but the first antenna **110** is not limited thereto. The first antenna **110** is located on a ceramic medium **114**, and located in the recess of an insulating support **150** and exposed to the insulating support **150**. The first antenna **110** has a feeding end **112**.
- (6) In the embodiment, the first antenna **110** is used for exciting at a first frequency band. The first frequency band is, for example, 1500 MHz to 1750 MHz, and the center frequency is, for example, 1575.42 MHz. The side length of the first antenna **110** is about $\frac{1}{2}$ wavelength of the first frequency band. In the embodiment, the resonance frequency of the first antenna **110** is not only determined by the side length of the first antenna **110**, but also related to the dielectric constant (ϵ_r) of the ceramic medium **114**.
- (7) The second antenna **120** is located on the upper surface of the insulating support **150**. The second antenna **120** is, for example, an LTE antenna, but the second antenna **120** is not limited thereto. The second antenna **120** includes a first radiator **121**, a second radiator **123**, and a third radiator **124**. The first radiator **121** is, for example, L-shaped and includes a feeding end **122**. The feeding end **122** is located at one end of the first radiator **121**. The first radiator **121** is used for exciting at a second frequency band. The second frequency band is, for example, 700 MHz to 960 MHz. In the embodiment, the length of the first radiator **121** is $\frac{1}{4}$ wavelength of the second frequency band.
- (8) The second radiator **123** is connected to the first radiator **121**. A part of the first radiator **121** (the section from the feeding end **122** to the junction between the first radiator **121** and the second radiator **123**) and the second radiator **123** may excite at a third frequency band. The third frequency band is, for example, 1710 MHz to 2170 MHz. The lengths of the part of the first radiator **121** (the section from the feeding end **122** to the junction between the first radiator **121** and the second radiator **123**) and the second radiator **123** are $\frac{1}{4}$ wavelength of the third frequency band.
- (9) The third radiator **124** is, for example, L-shaped. The third radiator **124** is connected to the first radiator **121** and includes a ground end **125** at the end. In the embodiment, the feeding end **122** and the ground end **125** of the second antenna **120** extend to the same side of the insulating support **150** (the right side of FIG. 1).
- (10) In the embodiment, the first antenna **110** is, for example, a rectangular microstrip antenna (a patch antenna), and the second antenna **120** is, for example, an inverted-F antenna (a PIFA antenna), but in other embodiments, either one of the first antenna **110** and the second antenna **120** may be a PIFA antenna, a monopole antenna, a loop antenna, or a rectangular patch antenna, and the types of the first antenna **110** and the second antenna **120** are not limited thereto.

(11) It may be seen from FIG. 1 that the second antenna **120** is disposed beside the first antenna **110**, and the first antenna **110** and the second antenna **120** are quite close to each other, for example, with the distance less than 5 mm between the first antenna **110** and the second antenna **120**. In this state, the antenna module **100** of the embodiment has a special design, so that there is favorable isolation between the first antenna **110** and the second antenna **120**, which will be described below.

(12) FIG. 2 is a schematic diagram of the antenna module shown in FIG. 1 with the insulating support, the first antenna, and the second antenna hidden. Referring to FIG. 1 and FIG. 2, the first ground plane **130** and the second ground plane **132** are located on the lower surface of the insulating support **150** (FIG. 1). In the embodiment, the first ground plane **130** is located below the first antenna **110** and spaced apart from the first antenna **110**. Specifically, the first antenna **110** and the first ground plane **130** are separated by the ceramic medium **114**. The size of the first ground plane **130** is greater than or equal to the size of the first antenna **110**, and the position of the first ground plane **130** corresponds to the position of the first antenna **110**, so that the projection of the first antenna **110** to the plane where the first ground plane **130** is located lies within the range of the first ground plane **130**.

(13) In addition, the second ground plane **132** is located below the second antenna **120** and spaced apart from the first ground plane **130**. In the embodiment, the ground end **125** of the second antenna **120** (FIG. 1) is connected to the second ground plane **132** through a conductor (not shown, such as a through hole).

(14) In addition, it may be seen from FIG. 1 and FIG. 2 that the size of the second ground plane **132** is greater than or equal to the size of the second antenna **120**, and the position of the second ground plane **132** corresponds to the position of the second antenna **120**, and the projection of the second antenna **120** to the plane where the second ground plane **132** is located lies within the range of the second ground plane **132**.

(15) The antenna module **100** of the embodiment may be applied to a bicycle tracker (tracker). Since the volume of the bicycle tracker is limited, the first antenna **110** and the second antenna **120** might be disposed very closely. In order to save space, it may be seen from FIG. 1 that the second antenna **120** surrounds the multiple sides (three sides) of the first antenna **110**.

(16) Correspondingly, it may be seen from FIG. 2 that the second ground plane **132** surrounds the multiple sides (three sides) of the first ground plane **130**. The first ground plane **130** and the second ground plane **132** are separated from each other, and the distance D between the first ground plane **130** and the second ground plane **132** is between 0.5 mm and 1.5 mm, for example, 1 mm.

(17) In addition, in the embodiment, the first ground plane **130** and the second ground plane **132** are coplanar and may be formed on the same layer of the circuit board, thereby increasing convenience of manufacturing. In addition, since the second antenna **120** requires a larger ground area, the area of the second ground plane **132** is greater than the area of the first ground plane **130**.

(18) In the conventional structure, the two antennas share the same ground plane, so the isolation is not favorable between the two antennas that are very close to each other. Through the design of the first antenna **110** and the second antenna **120** corresponding to the first ground plane **130** and the second ground plane **132**, respectively, and the first ground plane **130** and the second ground plane **132** being separated from each other, the antenna module **100** of the embodiment may enable the first antenna **110** and the second antenna **120** to have a more favorable isolation.

(19) It is worth mentioning that, in the embodiment, the antenna module **100** further includes a conductor spacer **140** connected to the second ground plane **132** and extending upward from the second ground plane **132** toward the second antenna **120**. The conductor spacer **140** is located between the first antenna **110** and a part of the second antenna **120** and separates the first antenna **110** from the second antenna **120**. That is to say, the conductor spacer **140** spaces apart the first antenna **110** and the second antenna **120** without contacting with the first antenna **110** and the second antenna **120**, so as to further improve the isolation between the first antenna **110** and the

second antenna **120**.

(20) More precisely, it may be seen from FIG. **1** that a part of the first radiator **121** and the second radiator **123** of the second antenna **120** form a U-shape by surrounding, and the first antenna **110** is located inside the U-shape. The bottom section of the U-shape (that is, the narrower section of the first radiator **121**) is where the first antenna **110** is closest to the second antenna **120**. In the embodiment, the conductor spacer **140** is located between the first antenna **110** and the section where the second antenna **120** is closest to the first antenna **110** (that is, the narrower section of the first radiator **121**) so as to improve the isolation between the first antenna **110** and the second antenna **120**.

(21) In the embodiment, the conductor spacer **140** is in a shape of a vertical plate or sheet, and is disposed on one of the vertical walls of the U-shaped notch of the insulating support **150**. Certainly, in other embodiments, the conductor spacer **140** may also be U-shaped, and be disposed on three vertical walls of the U-shaped notch of the insulating support **150** to surround the three sides of the first antenna **110**.

(22) It should be noted that, in the embodiment, the antenna module **100** further includes a third ground plane **134** located below the second ground plane **132**. Specifically, the first ground plane **130** and the second ground plane **132** are disposed on the upper surface of a substrate **136**, and the third ground plane **134** is disposed on the lower surface of the substrate **136**.

(23) In the embodiment, the third ground plane **134** is located on the entire lower surface of the substrate **136**. The third ground plane **134** is connected to the second ground plane **132** through multiple via holes **145** to increase the ground area of the second antenna **120**. In addition, the first ground plane **130** is insulated from the third ground plane **134**, such that the design may also enhance the isolation between the first antenna **110** and the second antenna **120**.

(24) It should be noted that, in other embodiments, the via holes **145** may be uniformly disposed on the second ground plane **132** or disposed on the edge of the second ground plane **132**, and the position and the quantity of the via holes **145** are not limited thereto.

(25) FIG. **3** is a frequency-S11 relation diagram of the first antenna and the second antenna of the antenna module shown in FIG. **1**. Referring to FIG. **3**, the first antenna **110** has favorable performance at the first frequency band (e.g., 1500 MHz to 1750 MHz, e.g., 1575.42 MHz), the second antenna **120** has favorable performance at the second frequency band (e.g., 700 MHz to 960 MHz) and the third frequency band (e.g., 1710 MHz to 2170 MHz), and S11 of the three frequency bands are all lower than -4 dB.

(26) FIG. **4** is a frequency-isolation relation diagram between the conventional antenna and the antenna module shown in FIG. **1**. Referring to FIG. **4**, compared with the isolation between the two antennas of the conventional structure, the isolation between the first antenna **110** and the second antenna **120** of the antenna module **100** of the embodiment may be dropped from -6.5 dB to -13.5 dB, which is below -10.0 dB, at the operating frequency band of 1575.42 MHz. Therefore, the isolation between the first antenna **110** and the second antenna **120** of the antenna module **100** of the embodiment may have a favorable performance.

(27) To sum up, the antenna module of the disclosure has the first antenna for exciting at the first frequency band and the second antenna for exciting at the second frequency band and the third frequency band so as to provide multiple applications. In the conventional structure, if the two antennas are too close, the isolation is poor. In the antenna module of the disclosure, the first ground plane is located below the first antenna and spaced apart from the first antenna. The second ground plane is located below the second antenna and spaced apart from the first ground plane. The ground end of the second antenna is connected to the second ground plane. Through the design of the first antenna and the second antenna having the corresponding first ground plane and second ground plane separately, the antenna module of the disclosure may enable the first antenna and the second antenna to have more favorable isolation.

Claims

1. An antenna module, comprising: a first antenna, comprising a first feeding end, for exciting at a first frequency band; a second antenna, disposed beside the first antenna, and comprising a first radiator with a second feeding end, a second radiator connected to the first radiator, and a third radiator connected to the first radiator and comprising a ground end, wherein the first radiator excites at a second frequency band, and a part of the first radiator and the second radiator excite at a third frequency band; a first ground plane, located below the first antenna, and spaced apart from the first antenna by an insulating support; and a second ground plane, located below the second antenna, and spaced apart from the first ground plane by the insulating support, wherein the ground end is connected to the second ground plane.
 2. The antenna module according to claim 1, further comprising a conductor spacer, connected to the second ground plane and extending from the second ground plane toward the second antenna, wherein the conductor spacer is located between the first antenna and a part of the second antenna, and separates the first antenna from the second antenna.
 3. The antenna module according to claim 1, further comprising a third ground plane, located below the second ground plane, and connected to the second ground plane through a plurality of via holes.
 4. The antenna module according to claim 3, wherein the first ground plane is insulated from the third ground plane.
 5. The antenna module according to claim 1, wherein the first ground plane and the second ground plane are coplanar, and an area of the second ground plane is greater than an area of the first ground plane.
 6. The antenna module according to claim 1, wherein a distance between the first ground plane and the second ground plane is between 0.5 mm and 1.5 mm.
 7. The antenna module according to claim 1, wherein a projection of the first antenna to a plane where the first ground plane is located lies within a range of the first ground plane, and a projection of the second antenna to a plane where the second ground plane is located lies within a range of the second ground plane.
 8. The antenna module according to claim 1, wherein the second antenna surrounds a plurality of sides of the first antenna, and the second ground plane surrounds a plurality of sides of the first ground plane.
 9. The antenna module according to claim 1, wherein a length of the first radiator is $\frac{1}{4}$ wavelength of the second frequency band, and lengths of the part of the first radiator and the second radiator are $\frac{1}{4}$ wavelength of the third frequency band.
 10. The antenna module according to claim 1, wherein a side length of the first antenna is a $\frac{1}{2}$ wavelength of the first frequency band.
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